October 11, 2026

The Editor-in-Chief
Neurocomputing

Dear Editor,

Please consider our manuscript *"What Carries Deep-Noise Robustness in a Mamba-3 Block? Input-Independent Scans and Noise-Dependent Gating under Cross-Condition Shift"* for publication in *Neurocomputing* as a regular research article.

Type of contribution. The paper contributes to the understanding of neural network architectures rather than proposing a new model. Selective state-space models (Mamba and its successors) are widely credited with robustness because of their input-dependent selectivity. We test this attribution with a pre-specified freeze–remove–add ablation chain on a Mamba-3 block, using cross-condition bearing-vibration data under matched additive noise as a controlled testbed. Making the scan's dynamics and read/write coefficients input-independent leaves the block's advantage over a time-invariant SSM intact, on both transitions of one dataset and with independent training and evaluation noise; removing the multiplicative output gate costs most at the deepest noise level, a contribution that grows significantly with noise depth is shown only for the fully frozen block on the original transition, and a multiplicative gate exceeds an additive branch of identical size at $-6\,\mathrm{dB}$. Results obtained under an earlier, shared-noise protocol are reported separately as descriptive evidence. The study therefore speaks directly to the journal's interest in architectures and the analysis of network behaviour; the vibration data serve as a testbed, not as the object of the contribution.

Rigour and transparency. Decision thresholds were written before each grid was run; for the later grids the pre-specification was committed to a public repository before the first run. Reporting at the final epoch is primary, and all data, code and decision records are publicly available.

Related submission. A related manuscript by the corresponding author and a co-author, "When does a second vibration sensor help? A controlled case study of sensor configuration for deep-learning bearing fault diagnosis", is under consideration at *Measurement Science and Technology* (MST-140500, submitted 8 October 2026). It addresses sensor-configuration choices and shares the experimental platform, but makes no claims about the component attribution studied here; no figure, table or result is shared between the two manuscripts. An auxiliary kinematics-alignment loss that appeared in early versions of the model plays no role in the present conclusions; all attribution arms are trained with cross-entropy only.

The manuscript is not under consideration elsewhere, and all authors have approved the submission.

Generative-AI use is declared in the manuscript and in Appendix C.

Sincerely,

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